Friday, 2 January 2026 5:57 PM

Static and User Input

```
a=5
b=6
print(A+B)
```

→ return a string

```
a=input ('Enter a value:') ← 5
b= input ('Enter another value:') ← 6
print(A+B).
```

→ addition →
↳ concatenation.

↳ 56

```
def magic ( ).
    a = int (input ('Enter a value:')
    b = int (input ('Enter another value')
    print (A+B)
```

return → is it a function. ← fx

function call → trigger.

```
magic()
    a=6
    b=10
    a+b =16.
```

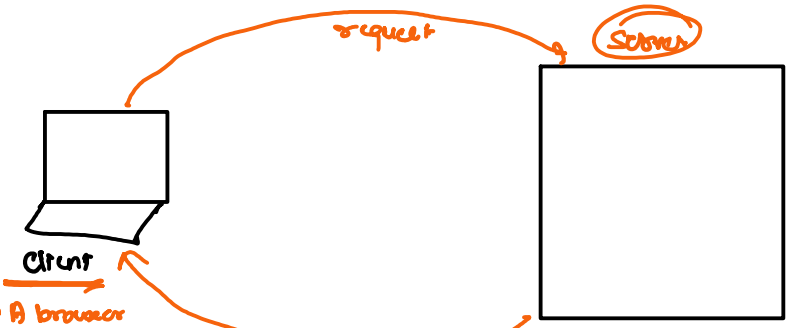
→ give me this value value back.

↳ • Module v/s package. → file exist in different folder

↳ random. →
math. →
os.
sys.
time.
etc.

numpy → random, math.

Introduce web Interface with flask.



- A browser
- mobile
- Postman.
- laptop
- tablet
- or any that can send a request

response

↳ html → type text Markup language

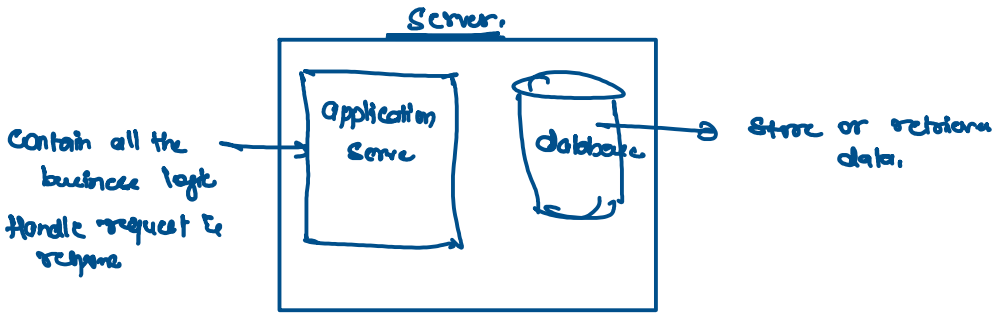
→ text → str.

- Type of request (HTTP methods).
  - GET → fetch data from the server
  - POST → Creating something on the server

- PUT → Update something on the server
- Delete → remove something from the server

Response Status Code

1xx → Informational
2xx → Success
3xx → Redirection
4xx → Client error
5xx → Server error
999 → customized response



How to create an application server in python?

→ Flask.
- → build application server
- → Define API's
- → Expose the functionality over the web.

The code written using flask is called
- Application code. / application program

4 steps mental model to create a flask application code.

Step-1 Import.
Step-2 Initialize flask object
Step-3. Create routes (Endpoints). → define where a request should go.
Step-4 run the application.

- Route (Endpoint) and function binding
  - Bind a route to a function mean
    - Whenever a request hits a specific route.
    - Flask automatically calls the functions written below it

So,
- route define where to go.
- Functions define what to do.

} → this binding is happening through decorator @app.route( )

- How flask automatically binds Routes to functions

whenever

@app.route("/") ?

def index

Flask automatically binds
the route /
To the function immediately written below it

browser

abc.com/,
abc.com/about

127.0.1.5000/ → Hello wer

- Default request type in flask.

If you do not explicitly specify a request method,
Flask listen to GET request by default.

@app.route('/').

- route matching.
  - For a request to be successful two thing must match
    - Route path. / /add}.
    - HTTP request
      - GET/POST must match what the route listens to.

Client

GET abc.com/ → listen GET → Yes (default).

POST abc.com/. → listen GET. → Error (HTTP 404/405)
→ exist → Yes

- Flask routes are strict about request type.
- GET & POST are not interchangeable.

- Explicitly allow post request

@app.route('/', method = ['POST']).

Now.

Route/ listen only to POST.
- GET. request → It will fail.

Route listen to POST
Client sent GET. } → Error.

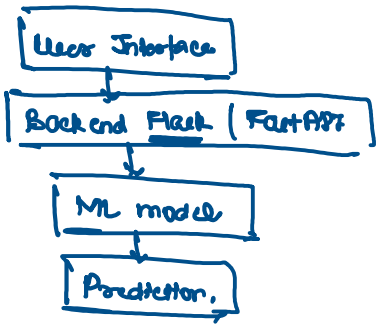
Route listen to GET
Client sent POST } → Error

Route path + request method must both match.

```
@ app.route('/', method = ['POST'] }
def...
```

```
@ app.route('/', method = ['GET'] } → overwritten.
def.
```

Real world data science system.



Youtube

Search → ML ranking model.

Recommendation
↳ Deep learning model

• Ola → prediction system.

ML engineer → build models.

Backend engineer → expose them.

You must understand how integration happens.

127.0.0.1:5000

• How do we send data from client to server dynamically?

Client → a = 6 and b = 10.

Query parameter

• Send the data via URL (Query parameter)

google.com/search?q=innomatics
↳ query parameter value.
↳ query parameter name.

127.0.0.1:5000/add?a=6&b=10.
↳ magic
↳ route.